

Alpha Diversity

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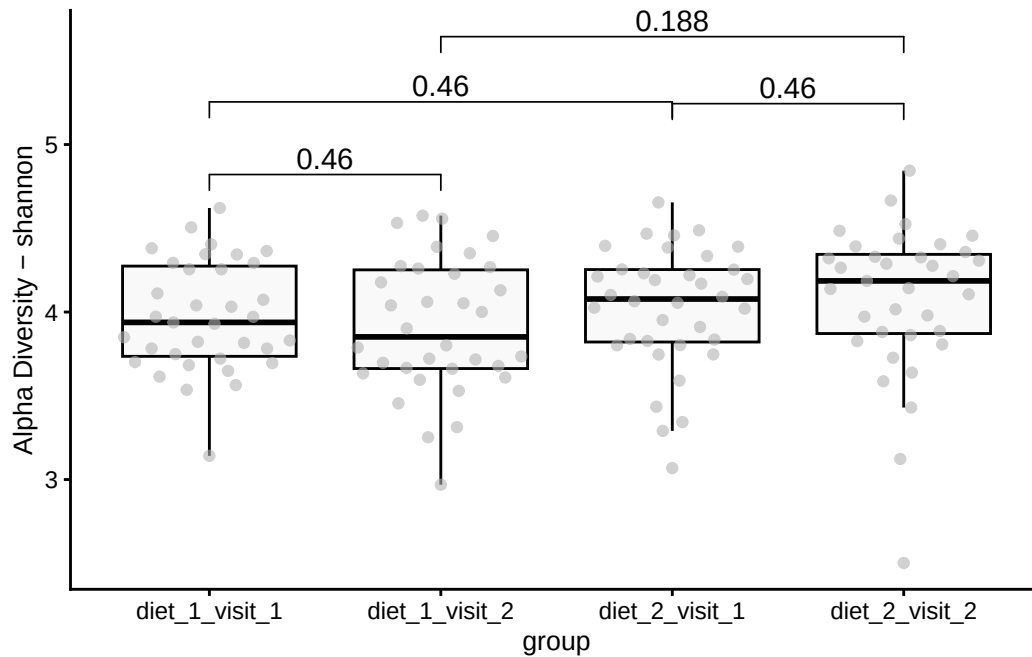
Introduction

This document reports on a microbiota project results of alpha diversity across different diets and visits.

Group-wise comparisons

- **Index:** shannon
- **Group variable:** group
- **Multiple testing correction:** fdr

Visualization



Characteristic	after			before		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	3.5	2.9, 4.1	<0.001	3.6	3.1, 4.0	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.28	-0.02, 0.58	0.065	0.20	-0.03, 0.44	0.085
3	0.20	-0.09, 0.50	0.2	0.11	-0.12, 0.35	0.3
4	0.10	-0.20, 0.39	0.5	0.17	-0.06, 0.40	0.2
age	0.01	0.00, 0.02	0.2	0.01	0.00, 0.01	0.2

¹CI = Confidence Interval

Paired test of before and after	Beta	95% CI ¹	p-value
(Intercept)	3.5	3.2, 3.9	<0.001
meal_group			0.11
1	—	—	
2	0.23	0.04, 0.42	0.017
3	0.16	-0.03, 0.34	0.094
4	0.13	-0.05, 0.31	0.2
age	0.01	0.00, 0.01	0.039

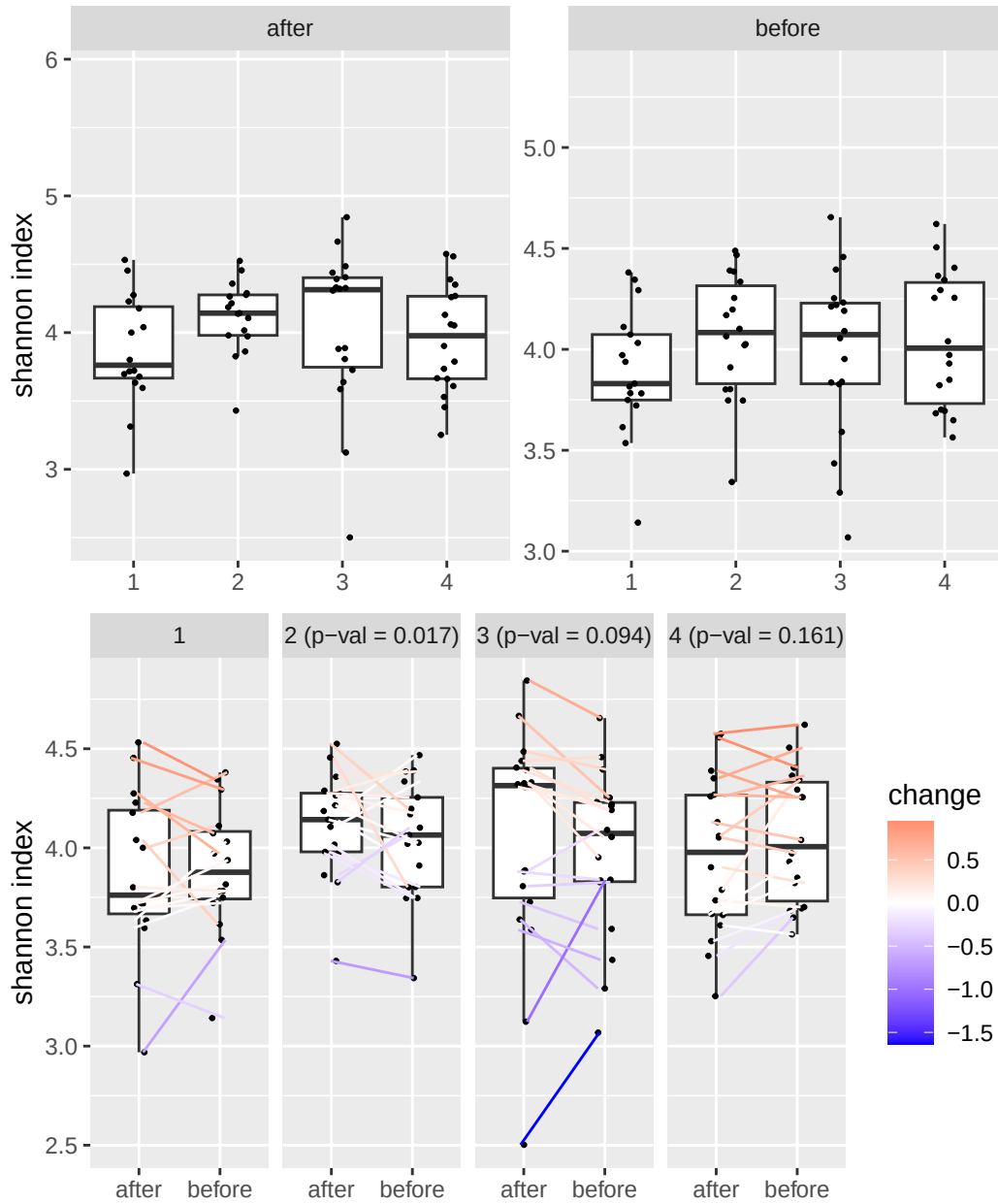
¹CI = Confidence Interval

Table of the top findings

Table 1: Table of alpha diversity comparisons

group1	group2	mean1	mean2	logFC	p.adj
diet_1_visit_1	diet_1_visit_2	3.97	3.91	-0.02	0.460
diet_2_visit_1	diet_2_visit_2	4.02	4.08	0.02	0.460
diet_1_visit_1	diet_2_visit_1	3.97	4.02	0.02	0.460
diet_1_visit_2	diet_2_visit_2	3.91	4.08	0.06	0.188

Alternate analyses



Data Summary

class: TreeSummarizedExperiment
dim: 1700 140

```

metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```

Version info

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R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS

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Matrix products: default
BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

```

```

locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

```

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time zone: Europe/Helsinki
tzcode source: system (glibc)

```

```

attached base packages:
[1] stats4      grid        stats      graphics  grDevices  utils      datasets
[8] methods    base

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other attached packages:
[1] kableExtra_1.4.0      vegan_2.6-8
[3] lattice_0.22-6        permute_0.9-7

```

[5]	lubridate_1.9.3	forcats_1.0.0
[7]	stringr_1.5.1	purrr_1.0.2
[9]	readr_2.1.5	tibble_3.2.1
[11]	tidyverse_2.0.0	tidyr_1.3.1
[13]	sechm_1.13.1	ComplexHeatmap_2.21.1
[15]	scater_1.33.4	scuttle_1.15.4
[17]	quarto_1.4.4	patchwork_1.3.0
[19]	parameters_0.24.0	multtest_2.61.0
[21]	miaTime_0.1.21	miaViz_1.15.4
[23]	ggraph_2.2.1	mia_1.15.6
[25]	TreeSummarizedExperiment_2.13.0	Biostrings_2.73.2
[27]	XVector_0.45.0	SingleCellExperiment_1.27.2
[29]	MultiAssayExperiment_1.31.5	SummarizedExperiment_1.35.3
[31]	Biobase_2.65.1	GenomicRanges_1.57.1
[33]	GenomeInfoDb_1.41.2	IRanges_2.39.2
[35]	S4Vectors_0.43.2	BiocGenerics_0.51.3
[37]	MatrixGenerics_1.17.0	matrixStats_1.4.1
[39]	gridExtra_2.3	gtsummary_2.0.4
[41]	ggsignif_0.6.4	ggpubr_0.6.0
[43]	ggplot2_3.5.1	DT_0.33
[45]	dplyr_1.1.4	cardx_0.2.2
[47]	ANCOMBC_2.8.0	

loaded via a namespace (and not attached):

[1]	randomcoloR_1.1.0.1	gld_2.6.6
[3]	nnet_7.3-19	TH.data_1.1-2
[5]	vctrs_0.6.5	energy_1.7-12
[7]	digest_0.6.37	png_0.1-8
[9]	shape_1.4.6.1	proxy_0.4-27
[11]	Exact_3.3	registry_0.5-1
[13]	rbiom_1.0.3	ggrepel_0.9.6
[15]	bayestestR_0.15.0	parallelly_1.38.0
[17]	mediation_4.5.0	MASS_7.3-61
[19]	reshape2_1.4.4	foreach_1.5.2
[21]	withr_3.0.2	xfun_0.49
[23]	ggfun_0.1.6	survival_3.7-0
[25]	doRNG_1.8.6	commonmark_1.9.1
[27]	memoise_2.0.1	ggbeeswarm_0.7.2
[29]	emmeans_1.10.4	systemfonts_1.1.0
[31]	gmp_0.7-5	tidytree_0.4.6
[33]	zoo_1.8-12	GlobalOptions_0.1.2
[35]	gtools_3.9.5	V8_5.0.1
[37]	Formula_1.2-5	datawizard_0.13.0

[39] httr_1.4.7	rstatix_0.7.2
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[43] rstudioapi_0.17.1	UCSC.utils_1.1.0
[45] generics_0.1.3	base64enc_0.1-3
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[51] polyclip_1.10-7	ca_0.71.1
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[63] readxl_1.4.3	magrittr_2.0.3
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[67] ggtree_3.13.1	DECIPHER_3.1.4
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[71] Hmisc_5.2-0	pillar_1.9.0
[73] nlme_3.1-165	iterators_1.0.14
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[93] multcomp_1.4-26	BiocSingular_1.21.4
[95] slam_0.1-54	e1071_1.7-16
[97] lmom_3.2	GetoptLong_1.0.5
[99] tidytext_0.4.2	splines_4.4.2
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[103] Rcpp_1.0.13-1	sparseMatrixStats_1.17.2
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[117] ggplotify_0.1.2	estimability_1.5.1
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[135]	cards_0.4.0	broom_1.0.7
[137]	coda_0.19-4.1	insight_1.0.0
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[141]	farver_2.1.2	tidygraph_1.3.1
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[159]	jsonlite_1.8.9	seriation_1.5.6
[161]	bit64_4.5.2	Rtsne_0.17
[163]	yulab.utils_0.1.8	BiocNeighbors_1.99.1
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[173]	boot_1.3-30	igraph_2.1.1
[175]	R6_2.5.1	broom.mixed_0.2.9.6
[177]	labeling_0.4.3	Rmpfr_1.0-0
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[193]	munsell_0.5.1	furrr_0.3.1
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